James McTighe

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EDUCATION

University of California, Berkeley **Graduation May 2026**
Masters of Molecular Science and Software Engineering *Berkeley, CA*

- Interdisciplinary curriculum blending core concepts in molecular science with software engineering principles, including data structures, algorithms, and computational methods.
- Machine learning & data science focus, with coursework in supervised/unsupervised learning, statistical modeling, and neural networks using Python libraries such as scikit-learn and TensorFlow.

Michigan State University **Graduation May 2019**
Bachelors of Science, Chemistry *East Lansing, MI*

- Summa Cum Laude ; 3.85/4.0 GPA
- Organic Synthesis research related to stereochemical Identification
  - Citation: Chakraborty, D., Zheng, L., Dai, Y., Gwasdacus, J., McTighe, J. E., Wulff, W. D., & Borhan, B. (2023). Employing a chiroptical sensor for the absolute stereochemical determination of α-amino and α-hydroxyphosphonates. *Chemical Communications*, *59*(84), 12629–12632. https://doi.org/10.1039/D3CC01757E

WORK EXPERIENCE

Orano Med **December 2024 – Present**
QC Analyst *Indianapolis, IN*

- Developed programs for molecular analysis based on physical properties of analytes
- Performed validation and qualification of analytical instruments in a controlled compliant setting.
- Created company program for outsourced testing of incoming goods
- Led project planning assessments which accelerated company goals / timelines

Cardinal Health **December 2021 – November 2024**
Senior Chemist *Indianapolis, IN*

- Created a Flask Based web application used for tracking equipment and materials at a site wide level.
- Performed release and stability testing of radiopharmaceutical drug products in a GMP-compliant environment, supporting tight manufacturing timelines
- Utilized HPLC, TLC, GC, gamma spectroscopy, and endotoxin testing to evaluate identity, purity, potency, and sterility in accordance with USP and FDA standards
- Executed and documented method development and optimization for new radiopharmaceutical formulations and analytical procedures
- Led and supported analytical method validation activities including accuracy, precision, linearity, specificity, and robustness studies
- Authored and reviewed validation protocols and reports, ensuring alignment with regulatory expectations and internal SOPs

Emergent BioSolutions **February 2021 – November 2021**
Data Analyst *Indianapolis, IN*

- Parsed and extracted structured data from technical SOPs to populate the LabVantage LIMS (Laboratory Information Management System), using ETL data techniques.
- Created scripts using Pandas to automatically parse incoming data and transform to match the global system.
- Applied logic-based business rules to standardize and transform site-specific data for alignment with global

enterprise systems, ensuring consistency across distributed platforms.

- Conducted peer reviews of team data inputs to validate adherence to system specifications and global data models, supporting data quality, governance, and schema compliance.

Midwest Compliance Labs **May2019 – January 2021**

Analytical Chemist *Terre Haute, IN*

- Conducted analysis of pharmaceutical stability samples utilizing chromatography methods / instruments
- Performed routine maintenance of HPLC and GC units.
- Assisted senior scientists in developing and validating LC-MS methods, enhancing throughput and reproducibility of metabolomics assays
- Prepared detailed technical reports and presentations to communicate findings and support project decisions
- Demonstrated strong understanding of organic and analytical chemistry principles to troubleshoot instrument issues and interpret complex datasets

PROJECTS

RNA Host Identification

- Developed a Convolution Neural Network / Transformer pipeline which predicted whether a virus could potentially mutate and infect a human host. Predicted with 97% accuracy against test cases

Natural Disaster Assessment

- Created a computer vision classification model using Scikit-Learn which evaluated both the type and severity of natural disasters. Predictions made with up to 96% Accuracy against test cases. The module includes hyperparameter tuning, cross-fold validation, and feature engineering.

Banking Database Management System Application

- Database Design & Modeling: Designed relational database schema with multiple entities such as accounts, customers, transactions, and loans. Developed entity-relationship diagrams (ERDs) and normalized the database to minimize redundancy and improve data integrity.
- Data Handling: Implemented complex SQL queries for transaction processing, account balance updates, and loan management, ensuring the accuracy and consistency of financial data.

Molecular Property Predictor

- Molecular Graph Construction: Created a graph representation of molecules with atoms as nodes and bonds as weighted edges, using the NetworkX library.
- Substructure Search: Implemented a method for searching and detecting specific substructures within a molecule using molecular fingerprints.

SKILLS

Python 3, C++, CMake, Bash Scripting, Git, GitHub, Scikit-learn, Scikit-image, PyTorch, Tensorflow, Pandas, NumPy, Seaborn, ArgParse, Linux Computing Systems, Computer Vision, Machine Learning, Deep Learning, Neural Networks, large-scale data processing, statistical modeling, optimization techniques, exploratory data analysis, graph based learning